# Notes on New Theoretical Directions for permABC

Working notes for discussion with co-authors

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**Abstract**

These notes collect a possible new theoretical angle for permABC, centered on its relation to Wasserstein ABC and on two asymptotic regimes that appear natural in the hierarchical setting: large number of observed compartments $K \to \infty$, and large over-sampling size $M \to \infty$. The goal is not to state definitive results yet, but to provide a clean framework for discussion, isolate plausible statements, and identify the points that should be formalized in future work.

## Contents

## 1 Motivation and purpose of these notes

The purpose of this short note is to present a possible extension of the current permABCstory in a form that can be discussed with co-authors before deciding whether any of it should enter the paper.

The main ideas are the following.

(1) permABCcan be viewed as a hierarchical analogue of WABCat the level of empirical distributions of *compartments*.

(2) This suggests a natural asymptotic regime when the number of compartments $K \to \infty$.

(3) The over-sampling bridge $M > K$ has a qualitatively different asymptotic behaviour: for large $M$, the global marginal tends to become prior-like, whereas the matched local variables become extreme-value / best-match objects, often nearly Dirac.

The point of these notes is therefore to organize these observations into a coherent narrative and make explicit which parts seem rigorous, which parts are currently heuristic, and which questions could realistically be turned into formal results.

## 2 Asymptotic links with Wasserstein ABC

In this section we clarify the relation between permABC and Wasserstein ABC (WABC), and we discuss two asymptotic regimes that seem especially relevant for our framework: (i) the number of observed compartments $K \to \infty$, and (ii) the over-sampling regime $M \to \infty$ with $K$ fixed.

Our goal is not to claim a complete theory at this stage, but rather to isolate the objects that should play the role of the WABC limits in our hierarchical setting, and to explain why the large-$M$ over-sampling posterior behaves very differently from the target permABC pseudo-posterior.

## 2.1 A compartment-level Wasserstein viewpoint

Recall that the observed data are

$$y = (y_1, \dots, y_K) \in \mathcal{D}^K,$$

where each compartment $y_k \in \mathcal{D}$ depends on a global parameter $\beta$ and a local parameter $\mu_k$. For simulated data

$$z = (z_1, \dots, z_K) \in \mathcal{D}^K,$$

our acceptance criterion is based on

$$\min_{\sigma \in S_K} d(y, z_\sigma).$$

It is useful to introduce the empirical measures on the space of compartments,

$$\widehat{\nu}_y^K := \frac{1}{K} \sum_{k=1}^{K} \delta_{y_k}, \qquad \widehat{\nu}_z^K := \frac{1}{K} \sum_{k=1}^{K} \delta_{z_k}.$$

If the compartment-level discrepancy is induced by a ground cost $\rho$ on $\mathcal{D}$, i.e.

$$d_K(y, z)^p := \frac{1}{K} \sum_{k=1}^{K} \rho(y_k, z_k)^p,$$

then

$$\min_{\sigma \in S_K} d_K(y, z_\sigma) = W_p(\widehat{\nu}_y^K, \widehat{\nu}_z^K),$$

where $W_p$ is the Wasserstein distance on $\mathcal{P}_p(\mathcal{D})$. If one uses instead an unnormalized distance, then the relation still holds up to the deterministic factor $K^{1/p}$.

Thus, once the data are viewed at the *compartment level*, PERMABC is the natural hierarchical analogue of WABC. The difference is that WABC works directly on an exchangeable sample $y_{1:K}$, whereas PERMABC also carries a latent representation

$$\theta = (\beta, \mu_1, \dots, \mu_K),$$

for which the compartment labels matter. The role of the projection step is precisely to recover an ordered local configuration after solving the transport/matching problem.

**Remark 1** (Dictionary with WABC)**.** *Ignoring the local labels, the permutation-invariant pseudo-posterior $\tilde{\pi}_\varepsilon$ behaves like WABC applied to the empirical distribution of compartments. By contrast, the projected pseudo-posterior $\pi_\varepsilon^\star$ contains extra information: it keeps track of which local parameters are attached to the optimally matched compartments.*

## 2.2 A WABC-style limit when $K \to \infty$

The relevant object when $K$ grows is not the full joint law of the labeled vector $(y_1, \dots, y_K)$, but the *marginal law of one compartment*. Define

$$\nu_\beta(\mathrm{d}y) := \int g(\mathrm{d}y \mid \beta, \mu)\, \pi_\mu(\mathrm{d}\mu \mid \beta),$$

where the conditional prior notation $\pi_\mu(\cdot \mid \beta)$ is only used for generality; in many examples it reduces to a fixed prior $\pi_\mu$.

Under the model, conditional on $\beta$, the compartments are exchangeable, and their empirical measure should converge to $\nu_\beta$. Likewise, if the observed data come from some true mechanism, we write $\nu_\star$ for the large-$K$ limit of $\widehat{\nu}_y^K$.

This suggests that the correct large-$K$ asymptotic target for the *global* parameter is

$$\beta^\star \in \arg\min_\beta W_p(\nu_\beta, \nu_\star),$$

exactly in the spirit of WABC.

**Proposition 1** (Heuristic WABC analogue for $K \to \infty$)**.** *Assume that:*

1. $\widehat{\nu}_y^K \Rightarrow \nu_\star$ *as* $K \to \infty$*;*
2. *for each fixed* $\beta$*, the simulated empirical measure satisfies* $\widehat{\nu}_z^K \Rightarrow \nu_\beta$ *in probability under the prior predictive model;*
3. *the map* $\beta \mapsto \nu_\beta$ *is identifiable in Wasserstein distance.*

*Then the permutation-invariant pseudo-posterior should behave as a coarsened posterior on* $\beta$*, namely*

$$\tilde{\pi}_{\varepsilon_K}(\mathrm{d}\beta \mid y) \approx \pi_\beta(\mathrm{d}\beta)\, \mathbf{1}\{W_p(\nu_\beta, \nu_\star) \lesssim \varepsilon_K\}\,,$$

*up to the empirical fluctuations of* $\widehat{\nu}_y^K$ *and* $\widehat{\nu}_z^K$*. In particular, if* $\varepsilon_K \to 0$ *slowly enough, one expects concentration near the set of minimizers of* $\beta \mapsto W_p(\nu_\beta, \nu_\star)$*.*

*Informal justification.* The acceptance event only depends on the optimally matched discrepancy between the two empirical measures of compartments. When $K$ is large, both empirical measures should be close to their population limits. Therefore the acceptance event becomes asymptotically equivalent to

$$W_p(\nu_\beta, \nu_\star) \lesssim \varepsilon_K,$$

which is exactly the WABC mechanism after replacing the original parameter $\theta$ by the integrated compartment law $\nu_\beta$. □

**Remark 2** (What this says — and what it does not say)**.** *This asymptotic picture is naturally a statement about* $\beta$*, not about the labeled vector* $\mu_{1:K}$*. The reason is that transport over compartments forgets labels. Therefore the WABC analogy is fundamentally a statement about the integrated model* $\beta \mapsto \nu_\beta$*.*

### 2.3 Why the local pseudo-posterior is not a usual Bayesian posterior

The previous discussion suggests a clean WABC-like limit for the global parameter, but not for the local parameters. This is not an accident.

Indeed, after permutation, the local parameters that appear in $\pi_\varepsilon^\star$ are not the original latent variables of the generative model in a labeled sense; rather, they are the local parameters *selected by the optimal matching.* Hence they should be interpreted as *transport witnesses* or *best local explanations* of the observed compartments, not as a standard posterior on pre-existing labeled latent variables.

For this reason, even when $K \to \infty$, the asymptotic behaviour of the local part of $\pi_\varepsilon^\star$ should be expected to differ from ordinary posterior consistency. At best, one might hope for convergence of the empirical law of matched locals, or for convergence toward a transport coupling between observed compartments and simulated latent explanations.

### 2.4 Large-$M$ over-sampling is a different asymptotic object

We now turn to over-sampling. Recall that for $M > K$ we simulate

$$\theta = (\beta, \mu_1, \ldots, \mu_M), \qquad z = (z_1, \ldots, z_M),$$

and accept if

$$\min_{\sigma \in \mathcal{A}_K^M} d(y, z_\sigma) \le \varepsilon,$$

where $\mathcal{A}_K^M$ denotes the set of injections from $\{1, \ldots, K\}$ into $\{1, \ldots, M\}$.

This defines two different objects:

- the *unprojected* over-sampling pseudo-posterior $\tilde{\pi}_\varepsilon^M$ on $(\beta, \mu_{1:M})$;
- the *projected* pseudo-posterior $\pi_\varepsilon^{\star,M}$ obtained after keeping the $K$ matched components selected by the optimal injection.

These two objects have very different large-$M$ limits.

**Proposition 2** (Heuristic large-$M$ behaviour of the unprojected law)**.** *Fix $K$ and $\varepsilon > 0$. Suppose that for a given $\beta$ and for every observed compartment $y_k$, the probability that one simulated compartment falls in an $\varepsilon$-matching region of $y_k$ is positive. Then the acceptance probability under over-sampling tends to one as $M \to \infty$. Consequently, the $\beta$-marginal of the unprojected pseudo-posterior satisfies*

$$\tilde{\pi}_\varepsilon^M(\mathrm{d}\beta \mid y) \longrightarrow \pi_\beta(\mathrm{d}\beta),$$

*or more generally to the prior restricted to the set of $\beta$ for which matching is feasible.*

*Informal justification.* For each $y_k$, let $B_{k,\varepsilon}(\beta)$ denote the event that one simulated compartment is close enough to $y_k$ to be part of a valid matching. Under independence across the $M$ simulated compartments, if each event has positive probability, then the probability that none of the $M$ candidates matches $y_k$ decays exponentially in $M$. Hence the probability that one can find $K$ distinct matches tends to one. Once acceptance becomes almost certain, the ABC filter no longer discriminates between values of $\beta$, so the marginal over $\beta$ reverts to its prior. □

This explains the phenomenon already observed numerically: when $M$ is very large, the unprojected over-sampling pseudo-posterior is not an approximation of the target PERMABC pseudo-posterior at all. It is instead an *auxiliary bridge* which becomes progressively less informative about $\beta$ as $M$ grows.

### 2.5 Why the projected local law becomes concentrated when $M \to \infty$

The projected law $\pi_\varepsilon^{\star,M}$ behaves very differently. Although the global marginal becomes almost prior-like, the selected local components are chosen among $M$ candidates by an extreme-value mechanism: they are the *best matches* among an increasingly large pool.

To formalize this, fix $(\beta, y_k)$ and let

$$c_\beta(y_k, \mu)$$

denote a local mismatch criterion. Depending on the context, one may take for example

$$c_\beta(y_k, \mu) = d(y_k, G(\beta, \mu))$$

in a deterministic local simulator $z = G(\beta, \mu)$, or an expected/marginal discrepancy in the stochastic case.

If the $\mu_j$ are i.i.d. from $\pi_\mu(\cdot \mid \beta)$, then the matched locals in the projected law are approximately the minimizers of $c_\beta(y_k, \mu_j)$ over $j = 1, \ldots, M$. Hence they are order-statistic-type variables.

**Proposition 3** (Heuristic concentration of projected locals as $M \to \infty$)**.** *Assume that for each $(\beta, y_k)$ the criterion $\mu \mapsto c_\beta(y_k, \mu)$ has a unique minimizer*

$$\mu_k^\dagger(\beta, y_k) \in \arg\min_\mu c_\beta(y_k, \mu),$$

*and that this minimizer lies in the support of the local prior. Then under the projected large-$M$ over-sampling law one expects*

$$\mu_k^{\star,M} \Longrightarrow \delta_{\mu_k^\dagger(\beta, y_k)} \qquad \textit{conditionally on } \beta,$$

*for each matched component $k = 1, \ldots, K$.*

In words: the selected local pseudo-posterior becomes increasingly concentrated around the best possible local explanation of each observed compartment under the current global parameter. This is why the large-$M$ local law can look "Dirac-like" even though the global marginal in $\beta$ stays essentially equal to the prior.

**Remark 3** (Deterministic and invertible local model)**.** *In the particularly transparent case where $z_k = G(\beta, \mu_k)$ deterministically and the map $\mu \mapsto G(\beta, \mu)$ is injective, the best local explanation is simply the inverse image of $y_k$, whenever it exists. In that case the large-$M$ projected local law really does converge to a Dirac mass at the inverse solution.*

### 2.6 A non-commutation of limits

The previous discussion highlights a key point:

$$\lim_{M\to\infty} \tilde{\pi}_\varepsilon^M(\mathrm{d}\beta \mid y) = \pi_\beta(\mathrm{d}\beta)$$

in general, whereas the target distribution is obtained at

$$M = K, \qquad \tilde{\pi}_\varepsilon^K = \tilde{\pi}_\varepsilon.$$

Therefore the limit $M \to \infty$ is *not* the inferential target. It is only useful because it creates an easier auxiliary distribution from which an SMC algorithm can start. This also explains why the schedule $M_0 > M_1 > \cdots > M_T = K$ is essential: the statistical meaning is only recovered at the end of the bridge.

### 2.7 Take-home messages

The discussion above suggests the following interpretation.

1. At the level of *empirical distributions of compartments*, PERMABC is the natural hierarchical analogue of WABC.
2. When $K \to \infty$, the relevant asymptotic object for the global parameter is the integrated compartment law $\nu_\beta$, and one expects WABC-style concentration toward minimizers of $\beta \mapsto W_p(\nu_\beta, \nu_\star)$.
3. This asymptotic picture does *not* imply ordinary posterior consistency for the labeled locals $\mu_{1:K}$, since the projection step turns them into matched witnesses rather than genuine labeled latent variables.
4. When $M \to \infty$ with $K$ fixed, over-sampling produces a qualitatively different object: the global marginal becomes prior-like, while the projected locals become extreme-value or best-match objects, often close to Dirac masses.
5. Consequently, over-sampling should be viewed as an algorithmic bridge, not as an inferential target in its own right.

### 2.8 Open questions

Several natural questions remain open.

- Can one prove a full WABC-style concentration result for $\beta$ under $K \to \infty$, with explicit rates?
- What is the correct asymptotic object for the empirical distribution of matched local parameters?

- What happens in a joint regime $K \to \infty$ and $M \to \infty$, for instance when $M/K \to c > 1$?
- Can the projected local limit be characterized as a transport map, a Monge coupling, or a nearest-neighbour selection rule?
- Which assumptions guarantee that the large-$M$ projected local law converges to a Dirac mass, and when does it instead converge to a non-degenerate set of minimizers?

We view these questions as promising directions for extending the current theoretical picture of PERMABC beyond the fixed-$K$ guarantees already established in this paper.

## 3 Practical discussion points for co-authors

At this stage, the most useful questions for discussion seem to be:

(1) Should the $K \to \infty$ discussion appear in the main paper, the supplement, or only in internal notes for now?

(2) Do we want to state the large-$K$ argument only for the global parameter $\beta$, leaving the local parameters $\mu_{1:K}$ explicitly outside the main theorem?

(3) Do we want to include a formal proposition explaining why large-$M$ over-sampling is not an inferential target, but only an algorithmic bridge?

(4) Would it be worth adding a simple toy example in which the large-$M$ behaviour can be shown analytically, e.g. a deterministic or Gaussian local model?

(5) Is there a clean way to phrase the projected local variables as transport witnesses, selected latent explanations, or approximate Monge matches?

## 4 Suggested next steps

A possible sequence of next steps would be:

(1) Clean the notation so that the WABC comparison uses exactly the same symbols as the current paper.

(2) Decide which parts are intended as theorem/proposition material and which parts should remain discussion/conjecture.

(3) Build one toy model where the $M \to \infty$ phenomenon is explicit.

(4) If the co-authors agree, turn the large-$K$ part into a concise proposition linked to the WABC literature.